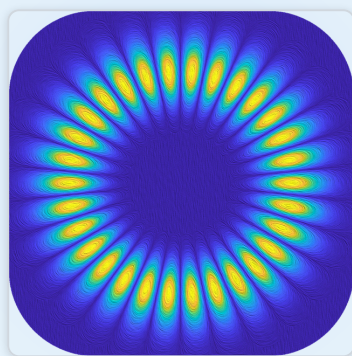




SteBeam

User Manual

A gift to everyone in optics

An ordinary industrial camera + SteBeam = a professional beam profiler

V0.1 Preview

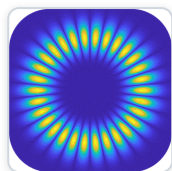
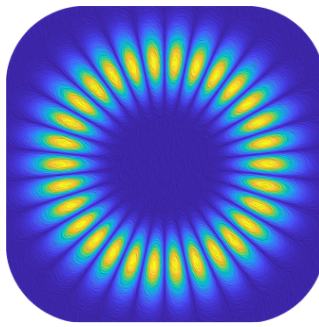


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SteBeam User Manual

SteBeam turns a plain industrial camera into a professional beam profiler.

This manual covers SteBeam V0.1 Preview: the interface is presented area by area, and every control's function, value range and usage requirements can be found in the matching section.

1. Overview

SteBeam is universal beam analysis software: connect an industrial area-scan camera and you get real-time beam imaging plus full quantitative measurement. Every frame from the camera is imaged and measured in real time, and presented in five views:

1. **2D Intensity** — pseudo-color intensity map with the measurement frame (ROI), centroid crosshair and 84% energy enclosure overlaid;
2. **3D Surface** — the beam rendered as a 3D intensity surface, rotatable by mouse drag, three resolutions, two shading modes;
3. **Beam Profile** — horizontal (X) and vertical (Y) intensity profiles through the centroid;
4. **Beam Size** — a sliding-window plot of beam diameter versus time;
5. **Centroid Track** — a drift-trajectory map of the beam position.

Measured live on every frame:

Category	Parameters
Size	Beam diameter (switchable: $D4\sigma$ / 84% encircled-energy / FWHM), ellipticity, major-axis orientation
Position	Centroid X/Y, peak X/Y, drift trajectory
Intensity	Power, peak intensity, peak saturation %FS

Key strengths:

1. **Standards-based**: second-moment beam widths per ISO 11146, with three diameter definitions for different engineering habits;

2. **Real-time:** full-frame acquisition with measurement and rendering completed on every frame — switch a view away and back, and not one frame of data is missing;
3. **Automatic background subtraction:** with the optional motorized shutter accessory, the “close shutter → sample dark background → subtract → reopen” sequence runs automatically; without a shutter, a manual occlusion path keeps every feature intact;
4. **Beam leveling:** one click rotates an elliptical spot’s major axis to vertical for clean comparison and reports;
5. **Data freedom:** every measurement exports to CSV in real time and beam images save with one click — your data is stored automatically and kept permanently.

2. System Requirements & Installation

Item	Requirement
OS	Windows 10 / 11 (64-bit)
Camera	Hikrobot MV-CU013-A0UM industrial area-scan camera (mono, 1280×1024 global shutter, USB)
Shutter accessory	Daheng GCI-7103M three-channel shutter driver box, optional; all measurement features work without it (see 9.3)

Supported devices

SteBeam V0.1 Preview currently supports these two devices (Figures 1 and 2):



Figure 1: Area-scan camera MV-CU013-A0UM

Hikrobot MV-CU013-A0UM industrial area-scan camera



Figure 2: Shutter driver box GCI-7103M

Daheng GCI-7103M three-channel shutter driver box (optional, for automatic background subtraction)

Two ways to install:

1. **Installer:** run the setup program and follow the wizard; a desktop shortcut can be created on request;
2. **Portable:** unzip anywhere and double-click `SteBeam.exe` — no installation needed.

On first launch without a camera attached, SteBeam reports the no-camera state but stays fully browsable.

3. Interface at a Glance

The main window has three areas (Figure 3):

1. **Top toolbar:** frame-source selection, start/stop acquisition, shutter, file saving, real-time storage, configuration, language switch (Section 5);
2. **Left view stack:** the five views above, with a frame-replay bar and view-switch buttons at the bottom (Section 6);
3. **Right settings panel** with three tabs —
 - **Home:** Controls (measurement conventions and display options) + Measures (all readouts) (Section 7);
 - **Setup:** six groups — Pseudo-Color Range, ROI, Calculation Threshold, Display, View Axes, Units and Photometric Calibration (Section 8);
 - **Acquisition:** four groups — Acquisition Params, Sensor Settings, Background Power, Statistics (Section 9).

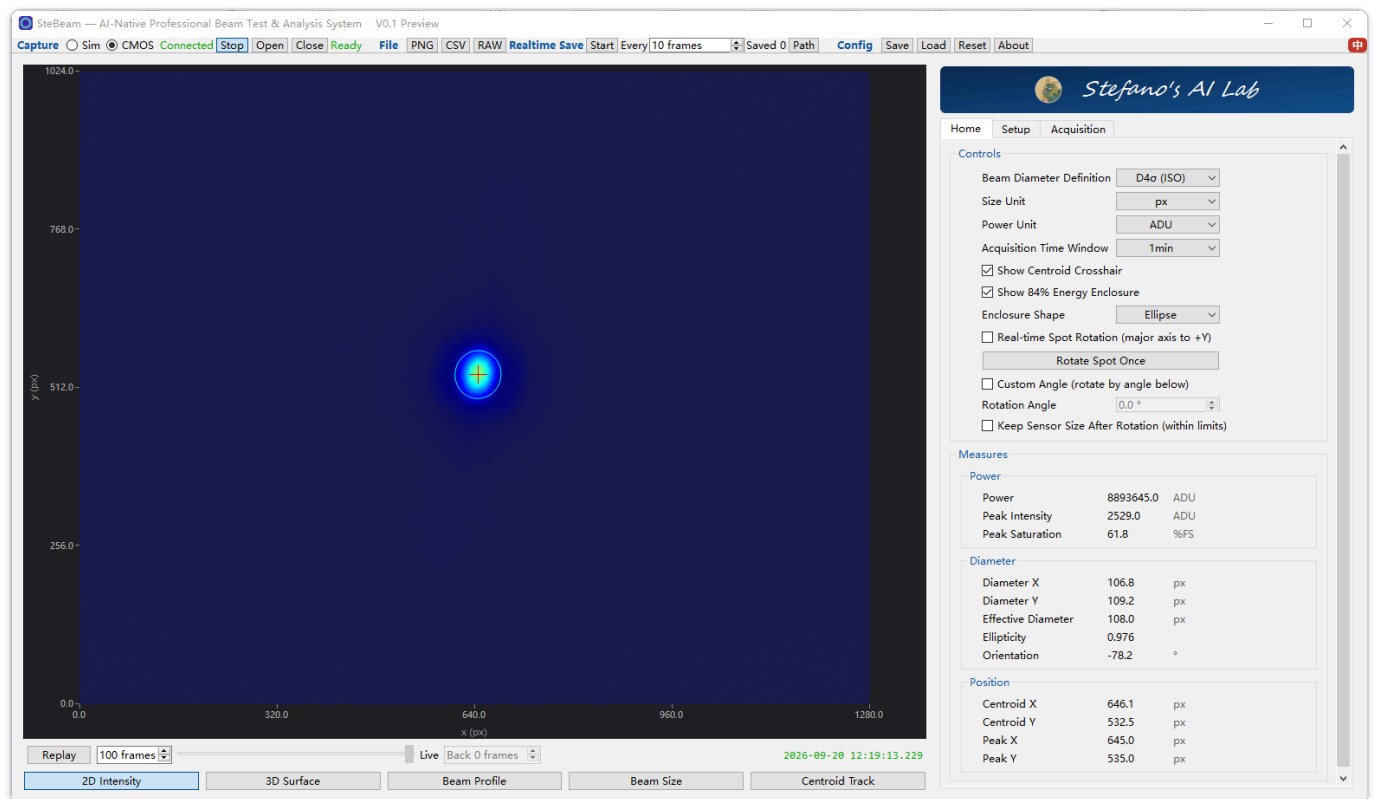


Figure 3: Main window

Main window (Home tab, live CMOS acquisition)

4. Quick Start

1. Connect the camera and launch SteBeam;
2. On the toolbar, make sure **CMOS** is selected as the frame source and the status reads "Connected", then use the acquisition controls to start — the live spot appears in the view area;
3. If the image is too bright or too dark, adjust **exposure** (preferred) or **gain** on the Acquisition tab, keeping **Peak Saturation** well away from 100 %FS;
4. On the Home tab, confirm the measurement conventions (diameter definition, size/power units); if needed, draw an **ROI** around the spot on the Setup tab and leave background subtraction on auto;
5. Read power, diameter and position on the Home tab — done.

5. Toolbar

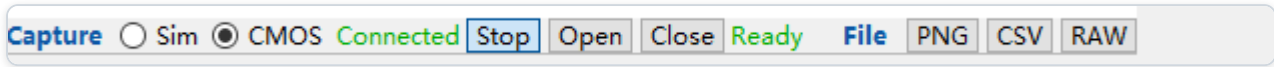


Figure 4: Toolbar, row 1

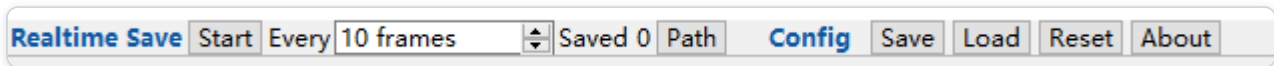


Figure 5: Toolbar, row 2

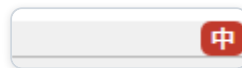


Figure 6: Language switch button

The toolbar runs left to right in four groups — Capture, File, Realtime Save, Config — with the language button at the far right (Figures 4–6).

5.1 Capture group

1. **Sim / CMOS (radio)**: frame-source selection. Sim generates a synthetic spot (demos and camera-less browsing); CMOS streams from the industrial camera. Note: switching takes effect immediately, no restart needed.
2. **Connection label (green text right of CMOS)**: camera connection status — “Connected” in green, with a matching label when not connected.
3. **Stop (checkable button)**: shown while acquiring; press to stop acquisition, toggle back to resume.
4. **Open / Close (buttons) + shutter status**: manual control of the optional motorized shutter accessory; the status label shows the live state (e.g. “Ready”). Note: the buttons are inert without the shutter accessory; the automatic background-sampling flow (see 9.3) drives the shutter itself — no manual intervention needed.

5.2 File group

All three buttons act on the **current view**; the default file name is the view title and the default location is “Documents”.

1. **PNG**: a what-you-see-is-what-you-get screenshot of the current view.
2. **CSV**: the numeric data of the current view — each of the five views has its own format (see 10.1).
3. **RAW**: the current frame’s raw pixels as a binary file (16-bit unsigned, row-major width×height, no header) for third-party analysis.

Note: all three require a valid current frame (acquisition started or a replay frame selected); in replay mode they save the selected replayed frame.

5.3 Realtime Save group

1. **Start (checkable button)**: turns automatic real-time storage on/off. Once on, 2D intensity data (CSV format with the metadata header block, see 10.2) is written to a folder at the set interval — no manual clicking.
2. **Every N frames**: the storage interval — one frame out of every N. Range: 1–1000, default 10.
3. **Saved N**: the number of files written in the current storage session.
4. **Path**: choose the destination folder. Default: `realtime_csv\` under the program directory, created automatically on first use.

Note: writing is asynchronous — if the disk writer is busy, frames are skipped rather than freezing the UI. Each frame becomes its own CSV file named after its capture time (`yyyyMMdd_HHmmss_zzz.csv`).

5.4 Config group

1. **Save / Load**: export all interface parameters to a settings file / restore them from one, via a file dialog — switch a whole configuration set between lasers or experiments in one click.
2. **Reset**: return every parameter to its factory default.
3. **About**: copyright, license and third-party notices.

5.5 Language button (far right)

The blue “En” / red “中” button (Figure 6) switches the UI between English and Chinese in one click; SteBeam restarts itself once and comes back entirely in the chosen language (see Section 11).

6. View Area: Five Views and Frame Replay

6.1 View-switch buttons

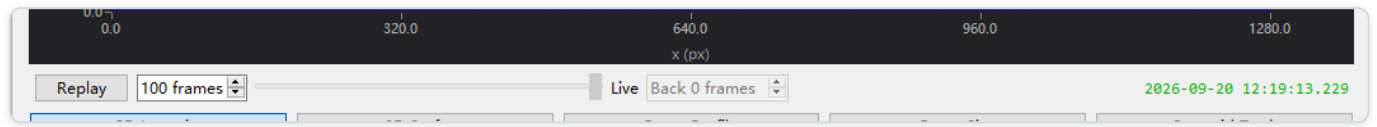


Figure 7: View switch and replay bar

Five buttons at the bottom of the view area map to the five views (Figure 7): **2D Intensity** / **3D Surface** / **Beam Profile** / **Beam Size** / **Centroid Track**. Switching is instant; all measurements run on every frame for all views at once, so switching away and back loses nothing.

6.2 The five views, photographed live

All shots below are direct CMOS captures (full-frame acquisition with the industrial camera).

2D Intensity: pseudo-color intensity map with the ROI frame, centroid crosshair and 84% energy enclosure overlaid; the bottom edge carries the display scale and axes (Figure 8):

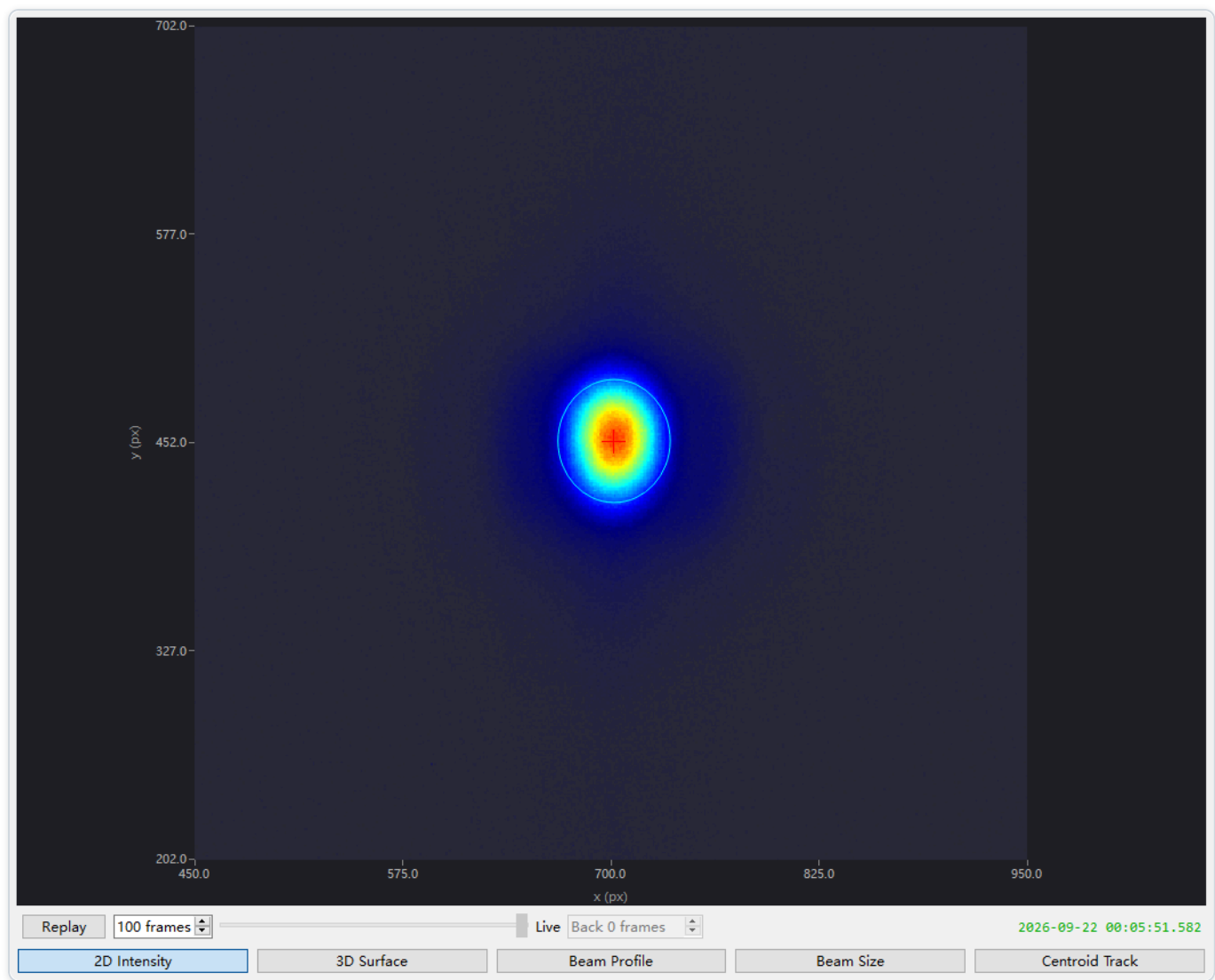


Figure 8: 2D Intensity

3D Surface: the same frame rendered as a 3D intensity surface — drag with the mouse to rotate the view (Figure 9):

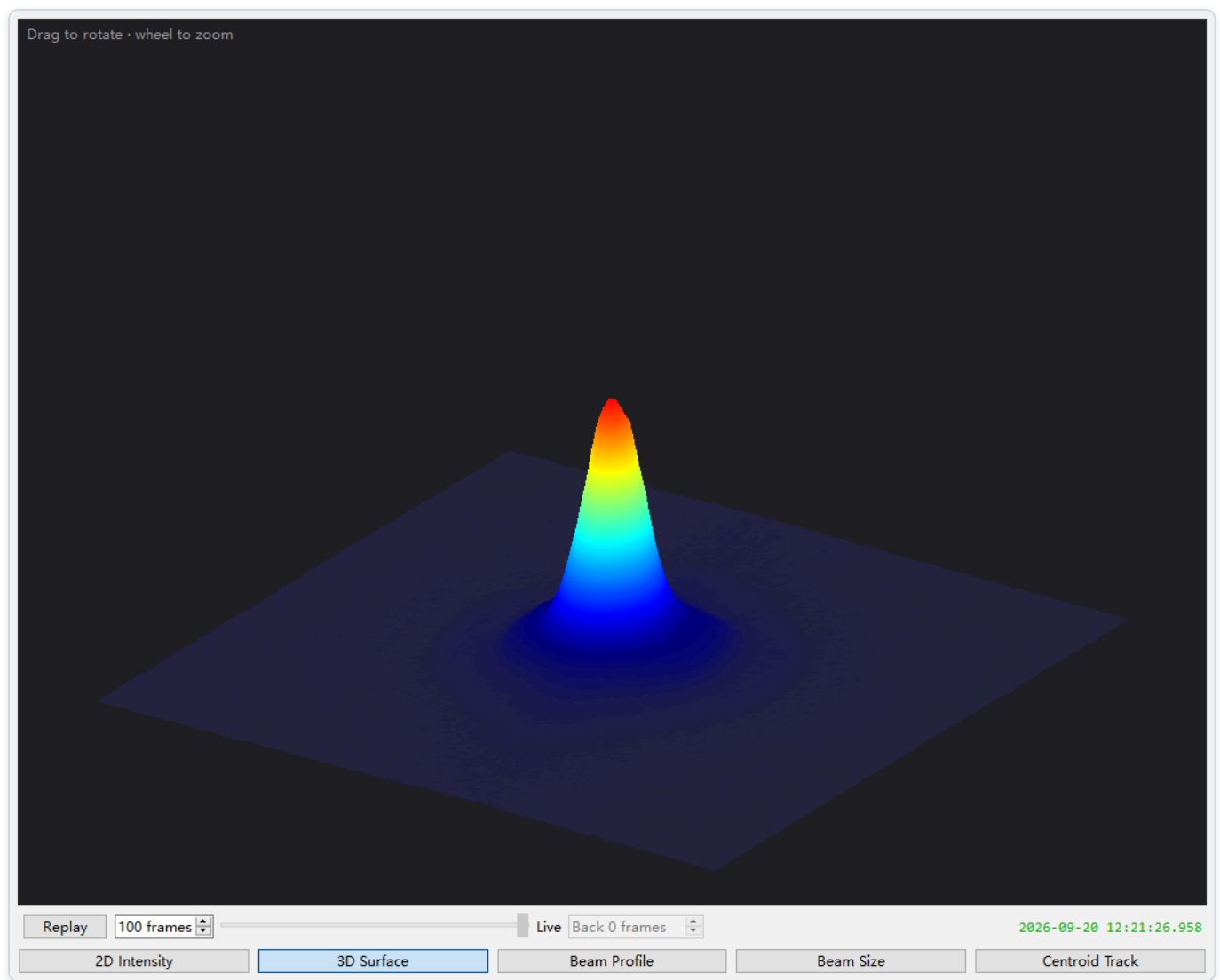


Figure 9: 3D Surface

Beam Profile: the horizontal (X) and vertical (Y) intensity curves through the centroid — use it to judge beam quality and spot sidelobe or double-peak structure (Figure 10):

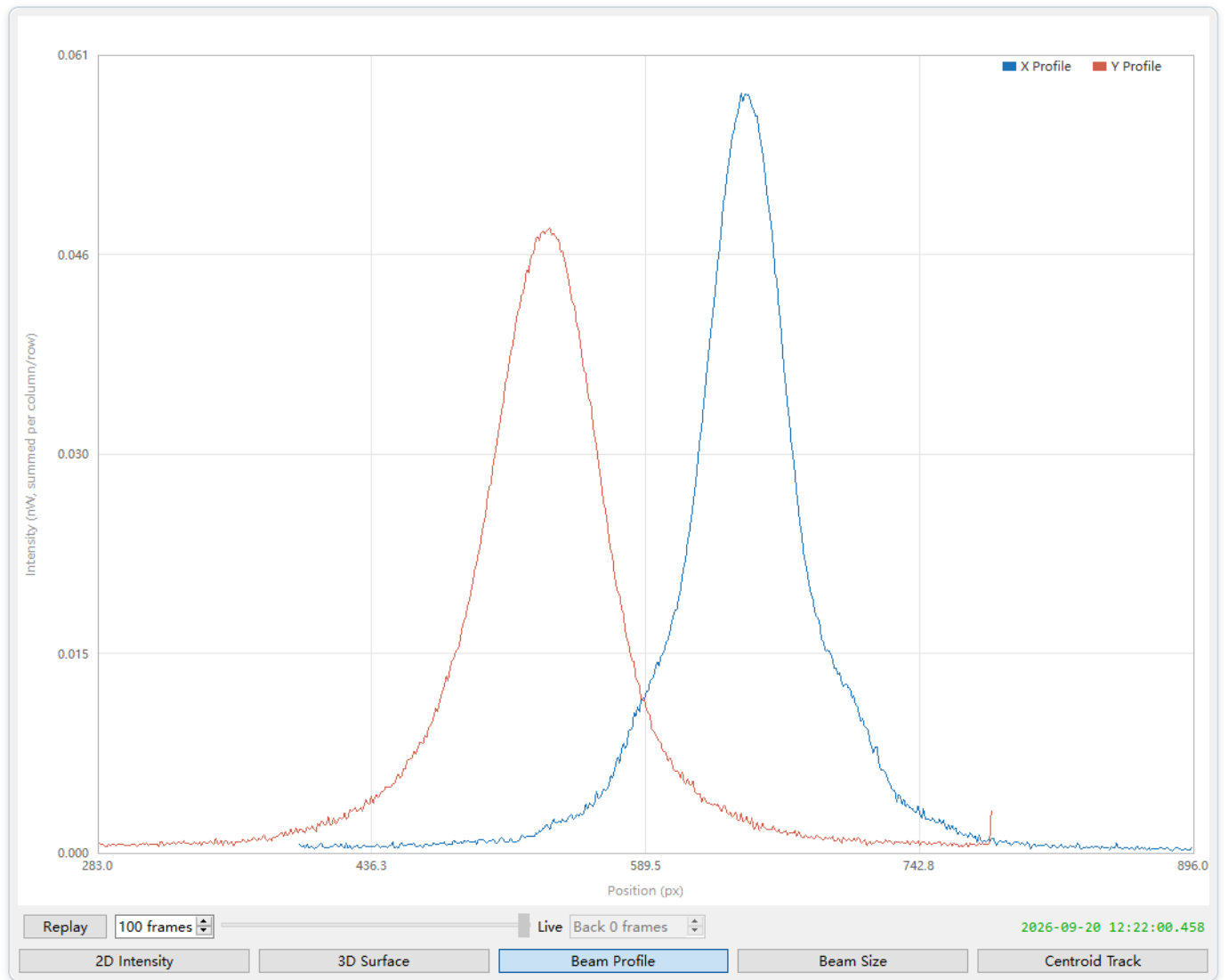


Figure 10: Beam Profile

Beam Size: sliding-window curves of the 84% encircled-energy width in X/Y over time — ideal for watching thermal lensing and mode jitter (Figure 11):



Figure 11: Beam Size

Centroid Track: a scatter map of historical centroid positions, darker dots meaning older — drift direction and amplitude at a glance. The view carries a **Clear track** button: it clears the recorded trajectory while new points keep recording (Figure 12):

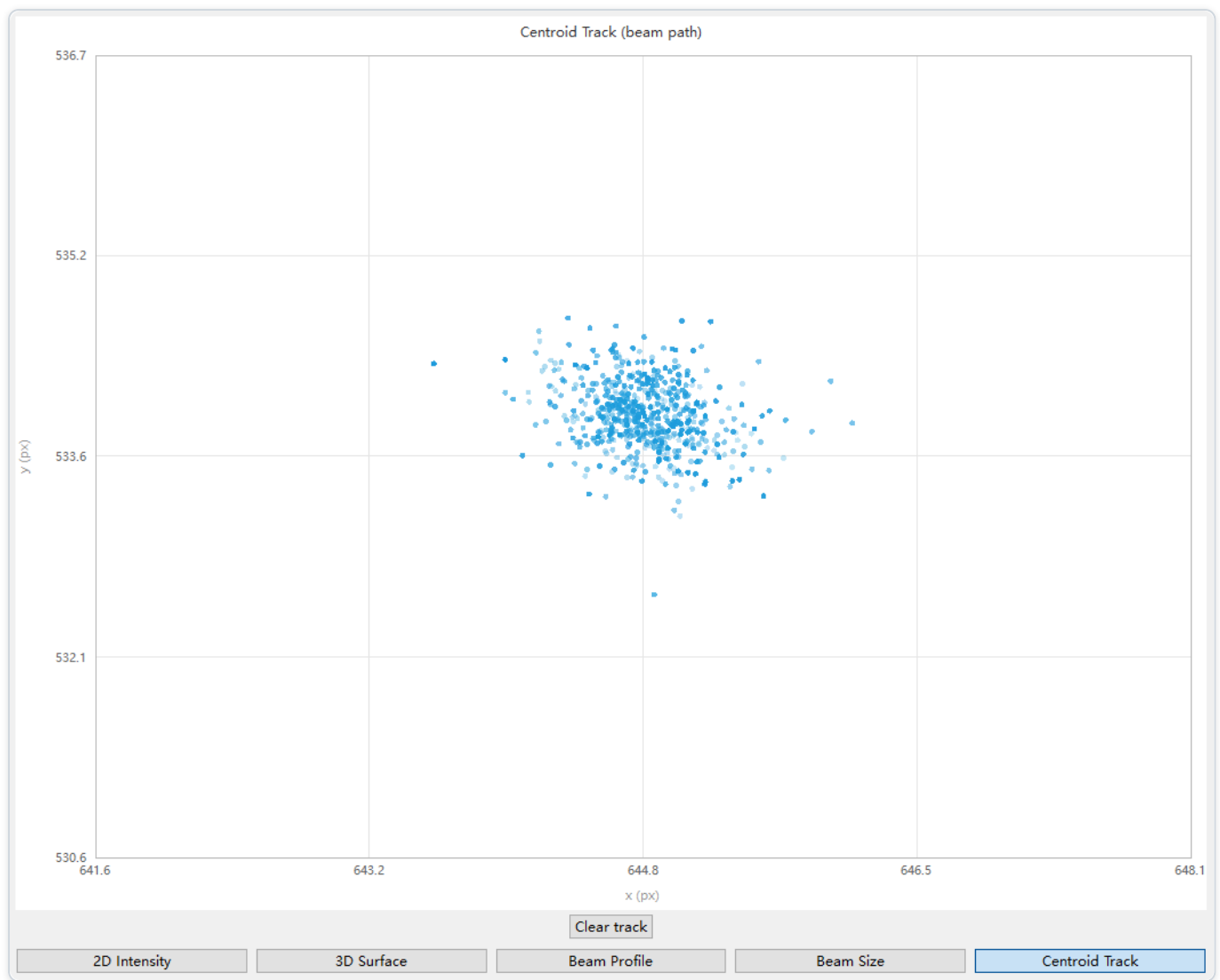


Figure 12: Centroid Track

6.3 Frame replay bar

Shared by the 2D Intensity, 3D Surface and Beam Profile pages (sitting just above the view-switch buttons; see Figure 7):

1. **Replay / Go live (checkable button)**: the single switch between replay and live mode. Pressing "Replay" stops acquisition automatically and freezes the buffer so you can scrub through history freely; pressing "Go live" resumes acquisition and returns to the live frame.
2. **Replay frame limit (spinbox)**: maximum frames kept in the replay buffer. Range: 0–100, default 100; 0 disables replay.
3. **Slider**: in replay mode, drag to pick a historical frame; the far right is the newest frame. Disabled in live mode (mode switching is by the "Go live" button only).
4. **Back N frames (spinbox)**: jump to a replay frame by number, in forward order: 1 = oldest, larger = newer; 0 = live.
5. **Timestamp (green text, right)**: capture time of the displayed frame, to the millisecond.

7. Home Tab: Controls and Measures

7.1 Controls group

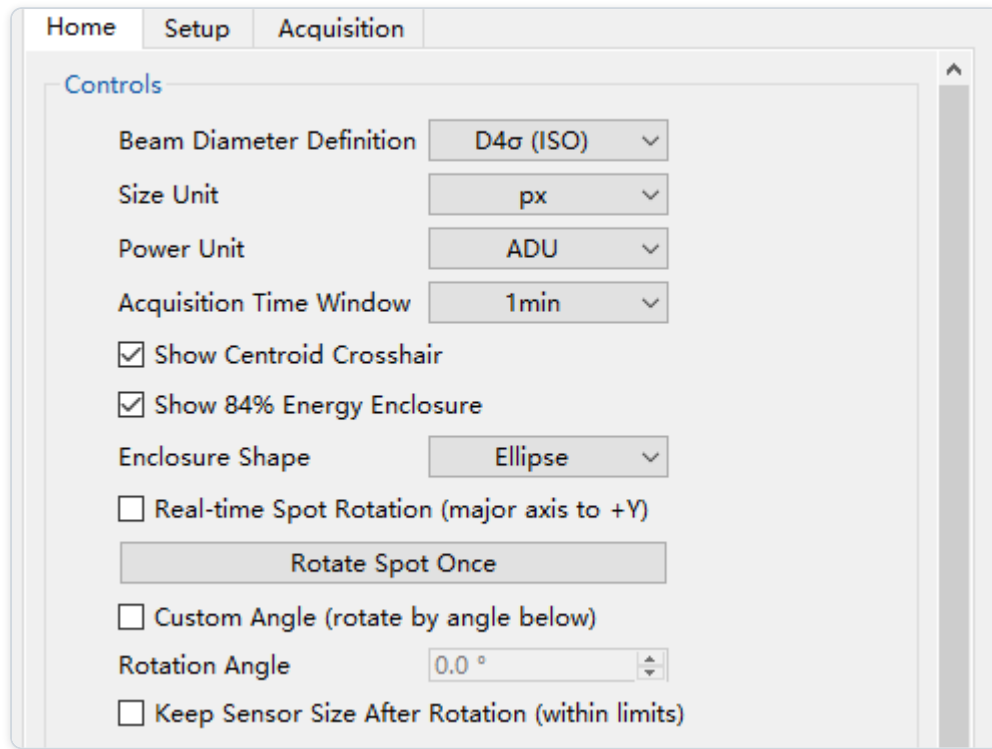


Figure 13: Home · Controls

The Controls group decides *what convention measures and displays* (Figure 13).

1. **Beam Diameter Definition (combo)**: pick one of three diameter definitions for the diameter readouts
 - **D4σ (ISO)**: ISO 11146 second-moment width (4× standard deviation) — the scientific convention, default;
 - **84% Energy**: width enclosing 84% of the energy — notably more robust against diffuse stray light;
 - **FWHM**: full width at half maximum — the classic engineering habit.
2. **Size Unit (combo)**: unit of every size readout (diameters, coordinates, curve axes) — px / μm / mm. px is the raw camera convention; μm/mm scale by the calibrated **Pixel Size** (see 8.6).
3. **Power Unit (combo)**: unit of the power and peak readouts — ADU / μW / mW / W. ADU is the raw camera reading; μW/mW/W go through the photometric calibration chain (see 8.6), and SteBeam picks the matching magnitude automatically.
4. **Acquisition Time Window (combo)**: sliding-window length for the Beam Size curves and Centroid Track — 1 / 5 / 10 / 30 / 60 min, default 1 min.
5. **Show Centroid Crosshair (checkbox)**: overlays the centroid crosshair on the 2D map, on by default.

6. **Show 84% Energy Enclosure (checkbox)**: overlays the 84% energy enclosure on the 2D map, on by default.
7. **Enclosure Shape (combo)**: shape of that enclosure — Ellipse / Rectangle, default Ellipse.
8. **Real-time Spot Rotation (major axis to +Y) (checkbox)**: every frame, the major-axis angle is auto-detected (50% peak gating) and the spot is rotated to put its major axis along +Y for display and measurement. Suits spots whose major axis slowly turns.
9. **Rotate Spot Once (button)**: at the moment of pressing, the major-axis angle is computed once from the current frame and that fixed angle is then applied continuously (it does not follow later frames). Mutually exclusive with Real-time Spot Rotation. Note: a fixed rotation keeps the output data-matrix size stable — handy for CSV batch analysis.
10. **Custom Angle (rotate by angle below) (checkbox)**: unlocks the Rotation Angle field and rotates by your entered angle, overriding auto-detection. Mutually exclusive with the two above.
11. **Rotation Angle (spinbox)**: range -180.0° to 180.0° , step 0.1° , positive = clockwise. Enabled only while Custom Angle is checked.
12. **Keep Sensor Size After Rotation (within limits) (checkbox)**: keeps the rotated output at sensor size (within limits), so rotation never changes the data-matrix dimensions.

Note (all three rotation modes): leveling only changes the display/measurement reference frame — all position readouts are converted back to true sensor coordinates, and tracking trajectories stay clean when the leveling angle changes.

7.2 Measures group

Measures		
Power		
Power	8893645.0	ADU
Peak Intensity	2529.0	ADU
Peak Saturation	61.8	%FS
Diameter		
Diameter X	106.8	px
Diameter Y	109.2	px
Effective Diameter	108.0	px
Ellipticity	0.976	
Orientation	-78.2	°
Position		
Centroid X	646.1	px
Centroid Y	532.5	px
Peak X	645.0	px
Peak Y	535.0	px

Figure 14: Home · Measures

The Measures group collects every live readout in three sub-groups: Power, Diameter, Position (Figure 14).

Coordinate convention (uniform across the whole program): position readouts use the sensor's bottom-left corner as origin with y pointing up — identical in the views, CSV and RAW.

1. **Power sub-group:**

- **Power:** background-subtracted total power inside the ROI. Unit follows the Power Unit combo: ADU is raw; $\mu\text{W}/\text{mW}/\text{W}$ come from the photometric calibration chain (factory constants are placeholders — calibrate absolute power with a single-point measurement, see 8.6 and Section 12).
- **Peak Intensity:** highest pixel intensity inside the ROI, same unit.
- **Peak Saturation:** peak intensity as a percentage of full scale (%FS). Usage: keep it well below 100 %FS — readings distort near full well (see Section 12).

2. **Diameter sub-group:**

- **Diameter X / Diameter Y:** horizontal/vertical beam widths under the current diameter definition (7.1 item 1); unit follows the Size Unit combo.
- **Effective Diameter:** $\sqrt{(\text{DiameterX} \times \text{DiameterY})}$ — a single number for spot size, convenient for trend curves.
- **Ellipticity:** minor/major axis ratio (1.0 = round) — inherently rotation-stable.
- **Orientation:** major-axis angle ($^\circ$) — inherently rotation-stable.

3. **Position sub-group:**

- **Centroid X / Centroid Y:** intensity-weighted first-moment position; unit follows the Size Unit combo.
- **Peak X / Peak Y:** peak-pixel coordinates.

8. Setup Tab: Analysis Parameters

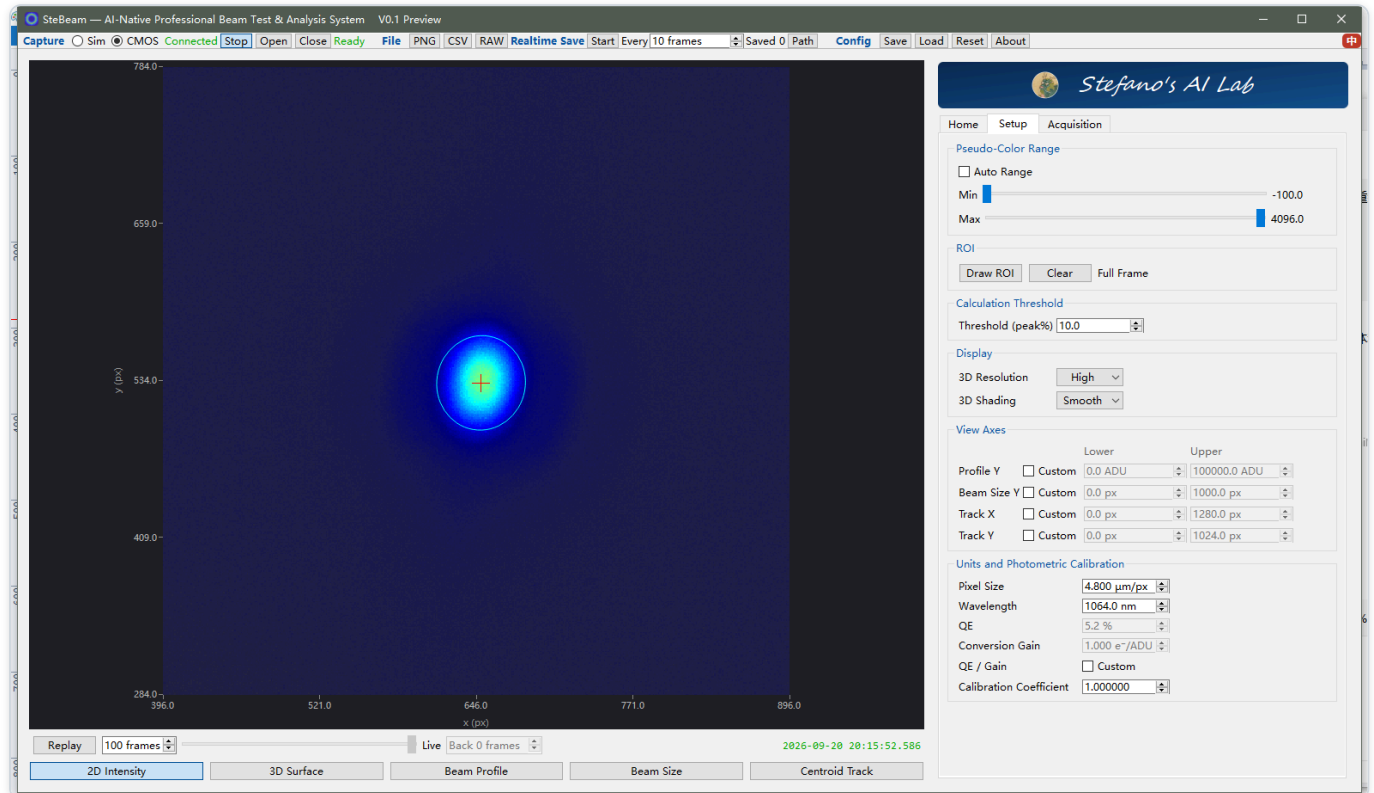


Figure 15: Setup tab, full window (live CMOS capture)

The Setup tab holds six groups top to bottom: Pseudo-Color Range, ROI, Calculation Threshold, Display, View Axes, Units and Photometric Calibration (Figures 15–21).

8.1 Pseudo-Color Range group

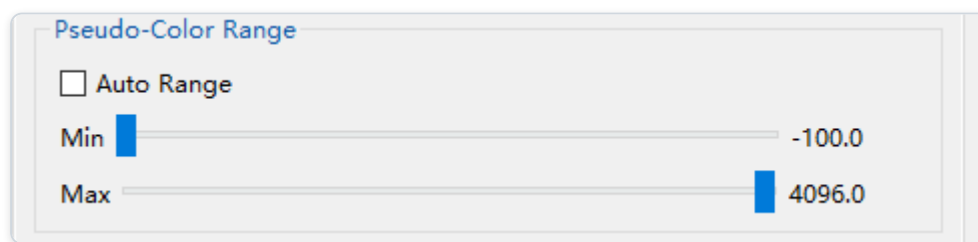


Figure 16: Pseudo-Color Range group

Sets the color-mapping range of the 2D map and 3D surface (display only — measurement values are untouched; Figure 16).

1. **Auto Range (checkbox):** when checked, the color scale adapts to every frame; off by default (manual mode).
2. **Min / Max (sliders):** manual color-scale lower and upper bounds. Range: -100 to 4096, step 16, default -100 / 4096.

8.2 ROI group

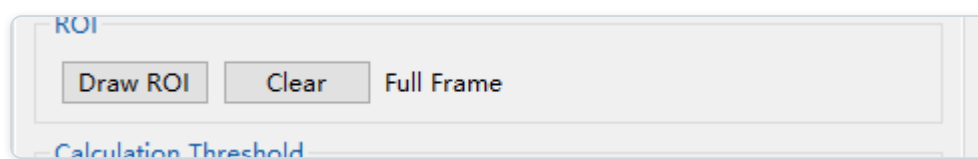


Figure 17: ROI group

The ROI (measurement frame) restricts where measurement happens — spots and stray light outside the frame are ignored (Figure 17).

1. **Draw ROI (button)**: enter draw mode and drag a rectangle on the 2D map.
2. **Clear (button)**: remove the ROI and measure the full frame again.
3. **Status label**: current ROI state; shows “Full Frame” by default.

8.3 Calculation Threshold group

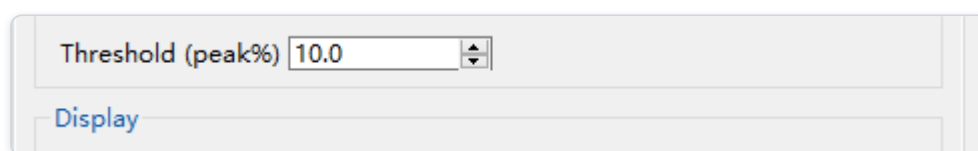


Figure 18: Calculation Threshold group

1. **Threshold (peak%) (spinbox)**: after background subtraction, pixels below (subtracted peak × threshold%) are excluded from width and centroid computation — the stray-light suppression control. Range: 0 to 50, step 0.1, default 10.0.
 - Physical cost: threshold gating slightly underestimates Gaussian beam widths (about 1.6× the threshold percentage). Measured calibration: at 5% the halo tail inflates widths by ~15%; at 10% the result matches the profile-method ground truth.
 - For a pure standards reading set 0 (trade-off: widths become more sensitive to background noise — pair it with proper background subtraction).

8.4 Display group

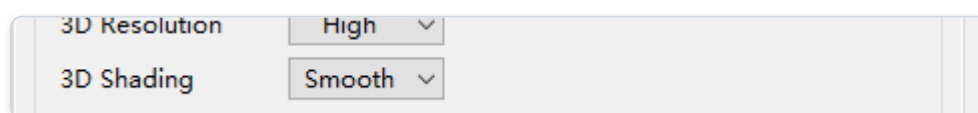


Figure 19: Display group

1. **3D Resolution (combo)**: 3D surface mesh density — Standard 96×76 / High 192×152 / Ultra 320×256, default High. Higher = crisper, more compute.
2. **3D Shading (combo)**: Fast = flat per-facet shading (fewer levels, cheaper); Smooth = per-pixel color interpolation (richer, natural transitions), default Smooth.

8.5 View Axes group

		Lower	Upper
Profile Y	☐ Custom	0.0 ADU	100000.0 ADU
Beam Size Y	☐ Custom	0.0 px	1000.0 px
Track X	☐ Custom	0.0 px	1280.0 px
Track Y	☐ Custom	0.0 px	1024.0 px

Figure 20: View Axes group

Manually set the numeric-axis limits of three views; with “Custom” unchecked the axis auto-ranges (default; Figure 20).

Row	View and axis	Default lower	Default upper	Value unit
Profile Y	profile vertical axis (intensity)	0	100000	follows Power Unit
Beam Size Y	size-curve vertical axis (diameter)	0	1000	follows Size Unit
Track X	track map horizontal axis	0	1280	follows Size Unit
Track Y	track map vertical axis	0	1024	follows Size Unit

1. **Custom (checkbox, one per row):** unlocks that row’s limit fields.
2. **Lower / Upper (spinboxes):** range -1e6 to 1e6; decimals follow the unit (1 for ADU/px, 3 for $\mu\text{m}/\text{mm}$ / $\mu\text{W}/\text{mW}/\text{W}$) and the suffix updates with the unit combo.

8.6 Units and Photometric Calibration group

Pixel Size	4.800 $\mu\text{m}/\text{px}$
Wavelength	1064.0 nm
QE	5.2 %
Conversion Gain	1.000 e^-/ADU
QE / Gain	☐ Custom
Calibration Coefficient	1.000000

Figure 21: Units and Photometric Calibration group

1. **Pixel Size (spinbox):** pixel pitch — converts pixels to real dimensions (all $\mu\text{m}/\text{mm}$ width and coordinate conversions ride on it). Range: 0.01–1000.000 $\mu\text{m}/\text{px}$, default 4.8 (the value confirmed in the MV-CU013-A0UM official datasheet).

2. **Wavelength (spinbox)**: current laser wavelength. Range: 100.0–20000.0 nm, default 1064 (just the common Nd:YAG line as a placeholder — set your actual wavelength). Note: the built-in QE curve is valid over 400–1100 nm and clamps to its endpoints outside.
3. **QE (spinbox)**: camera quantum efficiency at that wavelength (%). Defaults to the built-in official response-curve value ($\approx 5\%$ at 1064 nm; log-linear extrapolation beyond 940 nm — silicon band-edge cutoff makes 1064 nm a very weak parasitic response), locked by default; check “Custom” under QE/Gain to override.
4. **Conversion Gain (spinbox)**: electrons per ADU at 0 dB gain (e^-/ADU). Range: 0.001–1000.000; built-in 1.0 placeholder (the parameter is not published in official materials), locked by default; unlock via “Custom”. Note: analog gain in dB is folded into the conversion chain as $10^{(\text{dB}/20)}$.
5. **QE / Gain Custom (checkbox)**: off = QE from the built-in curve, gain = 1.0 placeholder; on = both fields unlocked for manual edit.
6. **Calibration Coefficient (spinbox)**: single-point calibration factor. Range: 0.0001–9999.000000, default 1.0. How to use: measure real power once with an external power meter, then fine-tune this coefficient until the reading matches — the power readout is then calibrated.

Power conversion chain: $\text{power} = \text{total ADU} \times \text{conversion gain} \div \text{QE} \times \text{single-photon energy (from wavelength)} \div \text{exposure time} \times \text{calibration coefficient}$. Factory constants are placeholder/built-in values — **absolute power readings are indicative only until calibrated**, while position, size and shape measurements are unaffected by calibration.

9. Acquisition Tab: Camera and Background Subtraction

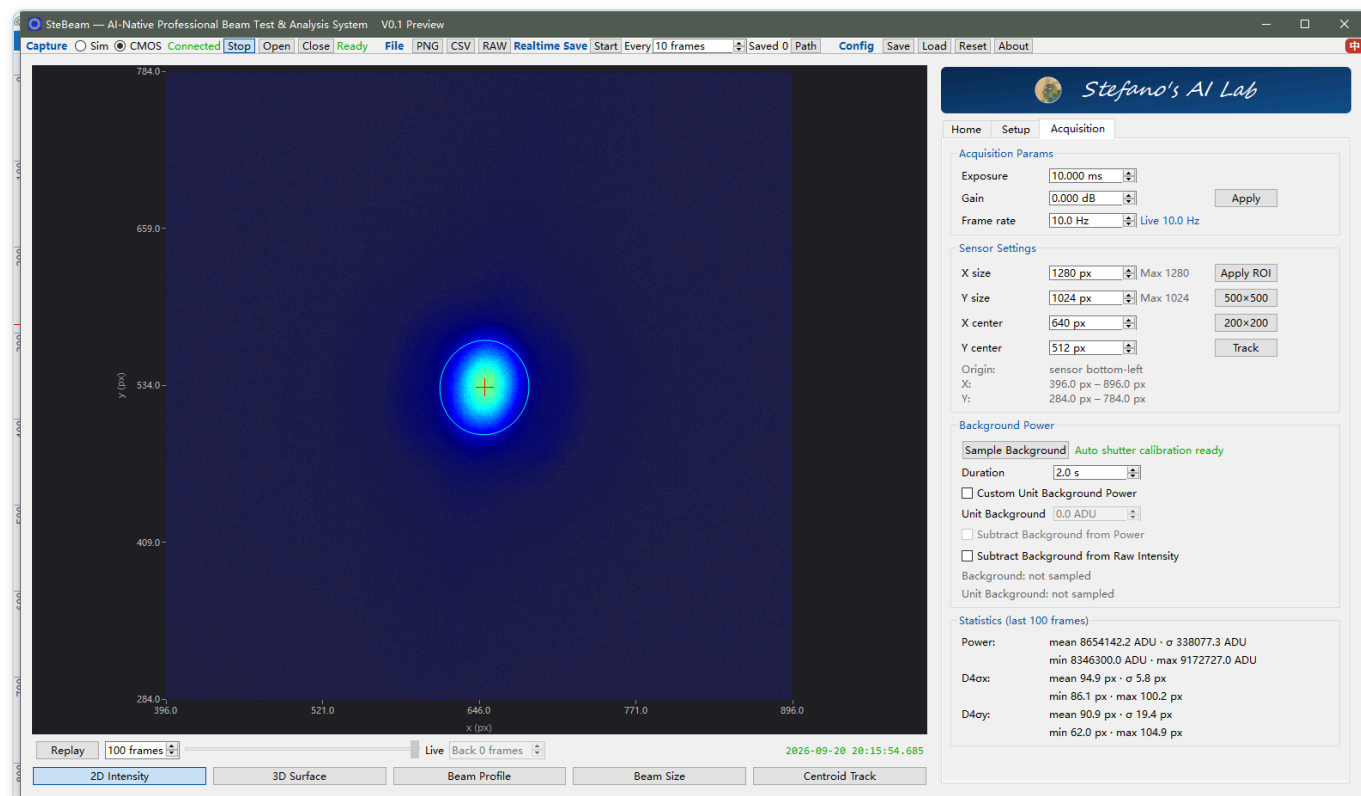


Figure 22: Acquisition tab, full window (live CMOS capture)

The Acquisition tab holds four groups top to bottom: Acquisition Params, Sensor Settings, Background Power, Statistics (Figures 22–26).

9.1 Acquisition Params group

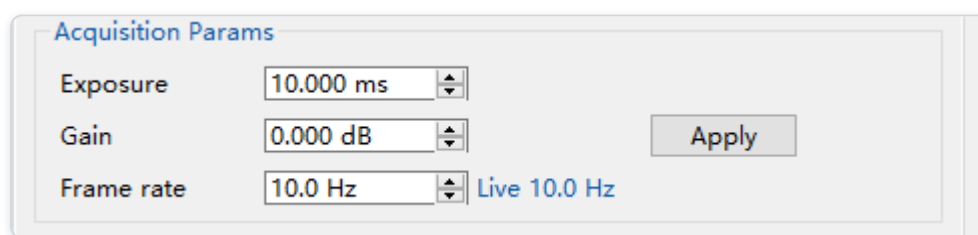


Figure 23: Acquisition Params group

1. **Exposure (spinbox)**: camera exposure time. Range: 0.009–10000.000 ms, default 10.000 ms. Usage: prefer exposure over gain for brightness (better noise behavior), and keep the peak unsaturated.
2. **Gain (spinbox)**: camera analog gain. Range: 0.000–24.000 dB, step 0.1.
3. **Frame rate (spinbox)**: target acquisition frame rate. Range: 1.0–240.0 Hz, step 1, default 10 Hz.
4. **Apply (button)**: sends exposure/gain/frame rate to the frame source. Note: all three are pushed automatically the moment they change; you can also press Apply to re-send manually.

5. **Live frame rate (blue text)**: measured frame rate, refreshed while streaming.

9.2 Sensor Settings group

Sensor Settings

X size	1280 px	Max 1280	Apply ROI
Y size	1024 px	Max 1024	500×500
X center	640 px		200×200
Y center	512 px		Track
Origin:	sensor bottom-left		
X:	396.0 px – 896.0 px		
Y:	284.0 px – 784.0 px		

Figure 24: Sensor Settings group

Crops the sensor window at acquisition time to cut data volume and raise the frame rate (the smaller the window, the higher the rate; Figure 24).

1. **X size (spinbox)**: range 4–1280 px, default 1280. Note: must be a multiple of 4.
2. **Y size (spinbox)**: range 4–1024 px, default 1024. Note: must be a multiple of 4.
3. **X center / Y center (spinboxes)**: window center coordinates, default 640 / 512 (sensor center). Ranges follow the window size (see the note lines under the group); origin = sensor bottom-left, y up.
4. **Apply ROI (button)**: sends the window size and center. Note: the camera briefly stops the stream to reconfigure — a short picture dropout is normal.
5. **500×500 / 200×200 (buttons)**: quick presets — set the window to that side length in one click, center unchanged, applied automatically.
6. **Track (button)**: copies the current beam centroid into the window centers and applies automatically. Note: a snapshot at press time, not live tracking — press again after the spot moves.
7. **Note lines (Origin / X / Y)**: the origin and extent of the window actually in effect, updated with your settings.

9.3 Background Power group

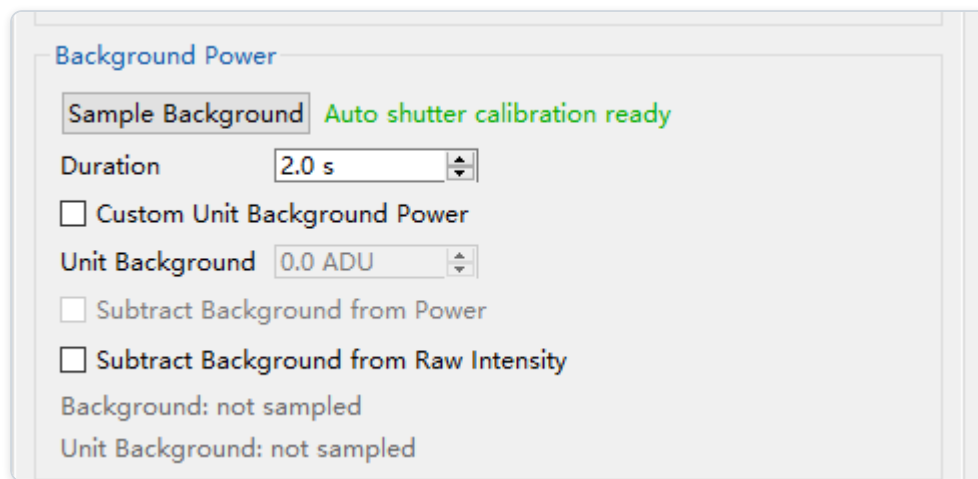


Figure 25: Background Power group

Stray light and detector dark current inflate readings; subtracting the background brings power and widths closer to the truth (Figure 25).

1. **Sample Background (button)**: samples the background floor. Two modes, matched automatically —
 - **Shutter connected (automatic)**: pressing it runs “close shutter → average the blocked-beam stray residual over the set duration → reopen shutter”; the shutter is always reopened when done;
 - **No shutter (manual)**: block the beam by hand first, then press. Note: an occlusion gate keeps the button disabled while the picture peak is still high (beam not fully blocked); the button is greyed out during sampling.
 - Always: sample the background at the **same exposure and gain** as the real beam, otherwise the subtraction is invalid; sampling needs the frame stream — if acquisition was stopped, it resumes automatically.
2. **Duration (spinbox)**: how long the background sample runs. Range: 0.5–60.0 s, step 0.1, default 2.0 s.
3. **Custom Unit Background Power (checkbox)**: off (default) = the per-frame pixel median is used as the unit background, immune to minute-scale background drift — right for almost every case; on = the manual “Unit Background” value below is used.
4. **Unit Background (spinbox)**: manual unit-background value. Range: 0–4095 ADU. Greyed out until “Custom” is checked.
5. **Subtract Background from Power (checkbox)**: whether the power readout subtracts the sampled background. Auto-checked and unlocked after a successful sample.
6. **Subtract Background from Raw Intensity (checkbox)**: when checked, raw intensity data (2D display, saved CSV, saved RAW) is output with the effective background already subtracted.
7. **Status lines (Background Power / Unit Background)**: two live readouts — background power (with the global power-unit conversion) and unit background; “not sampled” until a sample exists, with a hint if sampling fails (too few valid frames).

9.4 Statistics (last 100 frames) group

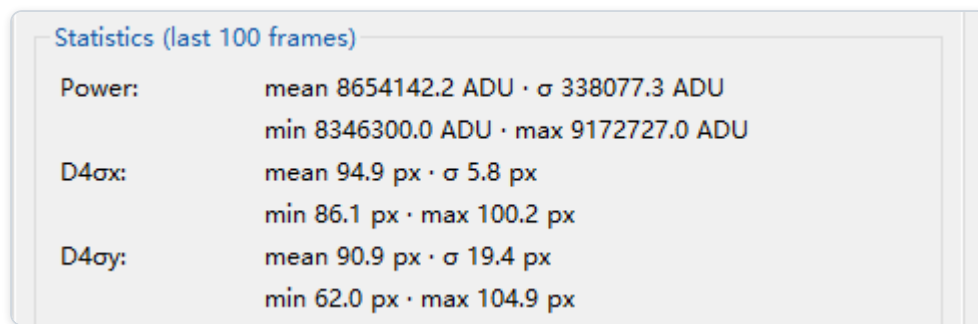


Figure 26: Statistics group

Rolling statistics over the last 100 frames, two lines per row: mean· σ and min·max (Figure 26).

1. **Power:** mean, standard deviation, minimum and maximum of the power readout;
2. **D4 σ x:** same four for the horizontal second-moment width;
3. **D4 σ y:** same four for the vertical second-moment width.

Units follow the current size/power unit combos. A quick read on short-term laser stability.

10. Saving Data & File Formats

10.1 Save CSV: five view-specific formats

The **CSV** button exports the **current view's** numeric data; the default file name is the view title.

1. **2D Intensity view**: metadata header block (6 lines, see 10.2) + the full-frame pixel matrix — one CSV row per image row, comma-separated, values as 12-bit ADU integers (background-subtracted values when "Subtract Background from Raw Intensity" is checked).
2. **3D Surface view**: a comment line `# 3D surface grid, W x H (bin-mean ADU)` + the down-sampled grid actually rendered by the view (bin means, row-major).
3. **Beam Profile view**: header `index, profileX, profileY` — one row per pixel position, the two through-centroid profile curves in raw ADU.
4. **Beam Size view**: header `t_min, d84X_<size unit>, d84Y_<size unit>` — the 84% encircled-energy width curves over the current acquisition time window (time in minutes).
5. **Centroid Track view**: header `x_px, y_px, timestamp` — every recorded centroid point (full-sensor pixel coordinates, bottom-left origin) + capture timestamp (`yyyy-MM-dd HH:mm:ss. zzz`).

10.2 2D intensity CSV metadata header block (same template for manual and real-time saves)

The header block is a fixed 6-line English block recording the full measurement context of the frame, so offline analysis can reproduce everything:

Line	Fields
1	Size Unit, Power Unit, Beam Diameter Definition (D4sigma (ISO) / 84% Energy / FWHM)
2	Background Subtracted / Background Not Subtracted, Background Power, Unit Background Power, Calculation Threshold (%)
3	Power, Peak Intensity
4	d_x, d_y (X/Y widths under the current diameter definition)
5	Pixel Size (um/px), Wavelength (nm), QE (%), Conversion Gain (e-/ADU), Calibration Coefficient (the five calibration parameters)
6	Exposure (ms), Gain (dB), Frame Rate (Hz), X Size (px), Y Size (px) (the six acquisition parameters; X/Y Size are the window sizes actually applied)

Unit convention: size and power units are declared only on line 1, and every later value shares that unit (power picks its magnitude automatically and then stays uniform file-wide, matching the on-screen readouts).

10.3 Real-time storage files

1. One CSV file per frame: metadata header block + pixel matrix;
2. file name = the frame's capture time `yyyyMMdd_HHmmss_zzz.csv` ;
3. folder = the directory chosen under **Path** (default `realtime_csv\` under the program directory);
4. interval = the **Every N frames** setting; if the writer is busy, frames are skipped rather than freezing the UI.

10.4 Save PNG and Save RAW

1. **PNG**: a what-you-see-is-what-you-get screenshot of the current view;
2. **RAW**: the current frame's raw pixels — 16-bit unsigned integers, row-major width×height, no header; parse with width = current acquisition X size, height = current acquisition Y size.

11. Language Switch

The language button at the far right of the toolbar toggles the UI between English and Chinese; SteBeam restarts itself once and comes back entirely in the chosen language. Saved files (CSV/PNG/RAW) are unaffected by the UI language — the CSV metadata header block is always the English template.

12. Usage Notes

1. **An honest note on long-wavelength power:** silicon cameras respond very weakly — and with large unit-to-unit variation — above roughly 1000 nm. For 1064 nm and similar wavelengths, position, size and shape measurements are fully reliable, but **absolute power** must be calibrated against an external power meter and the coefficient filled in (see 8.6);
2. **Avoid saturation:** readings distort as the peak approaches full well — lower the exposure or add attenuation;
3. **Threshold cost:** threshold gating slightly underestimates Gaussian beam widths (about 1.6× the threshold percentage); set 0 with proper background subtraction for strict standards readings (see 8.3);
4. **Background consistency:** always sample the background at the same exposure and gain as the real beam (see 9.3);
5. **Laser safety:** keep beam energy within the camera's tolerance and never let a high-power beam sit on the sensor.

13. License & Availability

SteBeam V0.1 Preview is a proprietary, free trial release; you may forward the full installer as-is. Trial terms are in the bundled `EULA.txt`. An open-source edition (LGPL-3.0) is coming soon. The bundled `LICENSES\` folder holds the license and attribution texts of the third-party open-source components.

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